



MATH230: Line Integrals and Vector Calculus

Introduction to Statistics · C-ID MATH230 · numberbender.com

Name: _____

Date: _____

Score: / 30

Learning Objectives

- Calculate the mean, median, mode, and range of a data set
- Compute sample variance and standard deviation
- Construct quartiles, the IQR, and identify outliers
- Describe the shape, center, and spread of a distribution

Simplify each expression completely. Show all steps and circle your final answer.

Flux and divergence

1. The divergence of a vector field is constant $\text{div } F = 2$ over a region $[0,4] \times [0,3]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 2 \, dA, [0, 4] \times [0, 3]$$

Answer: _____

2. The divergence of a vector field is constant $\text{div } F = 5$ over a region $[1,5] \times [1,2]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 5 \, dA, [1, 5] \times [1, 2]$$

Answer: _____

3. The divergence of a vector field is constant $\text{div } F = 6$ over a region $[0,4] \times [0,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 6 \, dA, [0, 4] \times [0, 5]$$

Answer: _____

4. The divergence of a vector field is constant $\text{div } F = 4$ over a region $[0,3] \times [0,4]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 4 \, dA, [0, 3] \times [0, 4]$$

Answer: _____



5. The divergence of a vector field is constant $\text{div } F = 3$ over a region $[1,4] \times [1,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 3 \, dA, [1, 4] \times [1, 5]$$

Answer: _____

6. The divergence of a vector field is constant $\text{div } F = 1$ over a region $[1,2] \times [0,4]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 1 \, dA, [1, 2] \times [0, 4]$$

Answer: _____

7. The divergence of a vector field is constant $\text{div } F = 5$ over a region $[1,3] \times [1,3]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 5 \, dA, [1, 3] \times [1, 3]$$

Answer: _____

8. The divergence of a vector field is constant $\text{div } F = 8$ over a region $[1,4] \times [1,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 8 \, dA, [1, 4] \times [1, 5]$$

Answer: _____

9. The divergence of a vector field is constant $\text{div } F = 2$ over a region $[1,5] \times [0,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 2 \, dA, [1, 5] \times [0, 5]$$

Answer: _____

10. The divergence of a vector field is constant $\text{div } F = 1$ over a region $[0,3] \times [0,4]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 1 \, dA, [0, 3] \times [0, 4]$$

Answer: _____

Green's theorem (flux form)



11. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,3] \times [1,4]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 3] \times [1, 4]$$

Answer: _____

12. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,2] \times [0,3]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 2] \times [0, 3]$$

Answer: _____

13. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,5] \times [0,4]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 5] \times [0, 4]$$

Answer: _____

14. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,5] \times [1,5]$. If the curl is constant = 3, find the circulation.

$$\iint_R 3 \, dA, [0, 5] \times [1, 5]$$

Answer: _____

15. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,3] \times [1,2]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [0, 3] \times [1, 2]$$

Answer: _____

16. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,3] \times [0,2]$. If the curl is constant = 5, find the circulation.

$$\iint_R 5 \, dA, [0, 3] \times [0, 2]$$

Answer: _____

17. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,3] \times [1,4]$. If the curl is constant = 4, find the circulation.

$$\iint_R 4 \, dA, [0, 3] \times [1, 4]$$

Answer: _____



18. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,5] \times [0,3]$. If the curl is constant = 4, find the circulation.

$$\iint_R 4 \, dA, [0, 5] \times [0, 3]$$

Answer: _____

19. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,5] \times [1,5]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [0, 5] \times [1, 5]$$

Answer: _____

20. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,5] \times [1,3]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 5] \times [1, 3]$$

Answer: _____

Work along a path

21. A vector field $F = (1, 3, 0)$ acts along a straight path with displacement vector $d = (2, 4, 0)$. Find the work done $W = F \cdot d$.

$$(1, 3, 0) \cdot (2, 4, 0)$$

Answer: _____

22. A vector field $F = (0, -4, 3)$ acts along a straight path with displacement vector $d = (5, 4, 2)$. Find the work done $W = F \cdot d$.

$$(0, -4, 3) \cdot (5, 4, 2)$$

Answer: _____

23. A vector field $F = (2, -1, 2)$ acts along a straight path with displacement vector $d = (6, 2, 2)$. Find the work done $W = F \cdot d$.

$$(2, -1, 2) \cdot (6, 2, 2)$$

Answer: _____

24. A vector field $F = (4, -2, 3)$ acts along a straight path with displacement vector $d = (5, 6, 1)$. Find the work done $W = F \cdot d$.

$$(4, -2, 3) \cdot (5, 6, 1)$$

Answer: _____



25. A vector field $F = (3, -5, 3)$ acts along a straight path with displacement vector $d = (6, 5, 1)$. Find the work done $W = F \cdot d$.

$$(3, -5, 3) \cdot (6, 5, 1)$$

Answer: _____

26. A vector field $F = (-2, -4, 3)$ acts along a straight path with displacement vector $d = (1, 2, 1)$. Find the work done $W = F \cdot d$.

$$(-2, -4, 3) \cdot (1, 2, 1)$$

Answer: _____

27. A vector field $F = (4, -3, 0)$ acts along a straight path with displacement vector $d = (2, 4, 1)$. Find the work done $W = F \cdot d$.

$$(4, -3, 0) \cdot (2, 4, 1)$$

Answer: _____

28. A vector field $F = (1, -1, 2)$ acts along a straight path with displacement vector $d = (5, 2, 1)$. Find the work done $W = F \cdot d$.

$$(1, -1, 2) \cdot (5, 2, 1)$$

Answer: _____

29. A vector field $F = (2, 3, 0)$ acts along a straight path with displacement vector $d = (5, 5, 0)$. Find the work done $W = F \cdot d$.

$$(2, 3, 0) \cdot (5, 5, 0)$$

Answer: _____

30. A vector field $F = (0, 3, 0)$ acts along a straight path with displacement vector $d = (3, 2, 0)$. Find the work done $W = F \cdot d$.

$$(0, 3, 0) \cdot (3, 2, 0)$$

Answer: _____





Topics: Green's theorem (flux form), Work along a path, Flux and divergence. All answers verified by independent computation.

Solutions



Flux and divergence

1. The divergence of a vector field is constant $\text{div } F = 2$ over a region $[0,4] \times [0,3]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 2 \, dA, [0, 4] \times [0, 3]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 2 \cdot (4-0) \cdot (3-0).$$

$$\rightarrow = 2 \cdot 4 \cdot 3 = 24.$$

Answer: $= 2 \cdot 4 \cdot 3 = 24$

2. The divergence of a vector field is constant $\text{div } F = 5$ over a region $[1,5] \times [1,2]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 5 \, dA, [1, 5] \times [1, 2]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 5 \cdot (5-1) \cdot (2-1).$$

$$\rightarrow = 5 \cdot 4 \cdot 1 = 20.$$

Answer: $= 5 \cdot 4 \cdot 1 = 20$

3. The divergence of a vector field is constant $\text{div } F = 6$ over a region $[0,4] \times [0,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 6 \, dA, [0, 4] \times [0, 5]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 6 \cdot (4-0) \cdot (5-0).$$

$$\rightarrow = 6 \cdot 4 \cdot 5 = 120.$$

Answer: $= 6 \cdot 4 \cdot 5 = 120$

4. The divergence of a vector field is constant $\text{div } F = 4$ over a region $[0,3] \times [0,4]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 4 \, dA, [0, 3] \times [0, 4]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 4 \cdot (3-0) \cdot (4-0).$$

$$\rightarrow = 4 \cdot 3 \cdot 4 = 48.$$

Answer: $= 4 \cdot 3 \cdot 4 = 48$

5. The divergence of a vector field is constant $\text{div } F = 3$ over a region $[1,4] \times [1,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 3 \, dA, [1, 4] \times [1, 5]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 3 \cdot (4-1) \cdot (5-1).$$

$$\rightarrow = 3 \cdot 3 \cdot 4 = 36.$$

Answer: $= 3 \cdot 3 \cdot 4 = 36$



6. The divergence of a vector field is constant $\text{div } F = 1$ over a region $[1,2] \times [0,4]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 1 \, dA, [1, 2] \times [0, 4]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 1 \cdot (2-1) \cdot (4-0).$$

$$\rightarrow = 1 \cdot 1 \cdot 4 = 4.$$

Answer: $= 1 \cdot 1 \cdot 4 = 4$

7. The divergence of a vector field is constant $\text{div } F = 5$ over a region $[1,3] \times [1,3]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 5 \, dA, [1, 3] \times [1, 3]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 5 \cdot (3-1) \cdot (3-1).$$

$$\rightarrow = 5 \cdot 2 \cdot 2 = 20.$$

Answer: $= 5 \cdot 2 \cdot 2 = 20$

8. The divergence of a vector field is constant $\text{div } F = 8$ over a region $[1,4] \times [1,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 8 \, dA, [1, 4] \times [1, 5]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 8 \cdot (4-1) \cdot (5-1).$$

$$\rightarrow = 8 \cdot 3 \cdot 4 = 96.$$

Answer: $= 8 \cdot 3 \cdot 4 = 96$

9. The divergence of a vector field is constant $\text{div } F = 2$ over a region $[1,5] \times [0,5]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 2 \, dA, [1, 5] \times [0, 5]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 2 \cdot (5-1) \cdot (5-0).$$

$$\rightarrow = 2 \cdot 4 \cdot 5 = 40.$$

Answer: $= 2 \cdot 4 \cdot 5 = 40$

10. The divergence of a vector field is constant $\text{div } F = 1$ over a region $[0,3] \times [0,4]$. By the Divergence Theorem (2D), find the net flux out of the region.

$$\iint_R 1 \, dA, [0, 3] \times [0, 4]$$

$$\rightarrow \text{Flux} = \text{integral integral } (\text{div } F) \, dA = 1 \cdot (3-0) \cdot (4-0).$$

$$\rightarrow = 1 \cdot 3 \cdot 4 = 12.$$

Answer: $= 1 \cdot 3 \cdot 4 = 12$



Green's theorem (flux form)

11. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,3] \times [1,4]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 3] \times [1, 4]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 1 \cdot (3-1) \cdot (4-1) = 1 \cdot 2 \cdot 3 = 6.$$

Answer: $= 1 \cdot 2 \cdot 3 = 6$

12. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,2] \times [0,3]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 2] \times [0, 3]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 1 \cdot (2-1) \cdot (3-0) = 1 \cdot 1 \cdot 3 = 3.$$

Answer: $= 1 \cdot 1 \cdot 3 = 3$

13. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,5] \times [0,4]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 5] \times [0, 4]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 1 \cdot (5-1) \cdot (4-0) = 1 \cdot 4 \cdot 4 = 16.$$

Answer: $= 1 \cdot 4 \cdot 4 = 16$

14. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,5] \times [1,5]$. If the curl is constant = 3, find the circulation.

$$\iint_R 3 \, dA, [0, 5] \times [1, 5]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 3 \cdot (5-0) \cdot (5-1) = 3 \cdot 5 \cdot 4 = 60.$$

Answer: $= 3 \cdot 5 \cdot 4 = 60$

15. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,3] \times [1,2]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [0, 3] \times [1, 2]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 1 \cdot (3-0) \cdot (2-1) = 1 \cdot 3 \cdot 1 = 3.$$

Answer: $= 1 \cdot 3 \cdot 1 = 3$



16. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,3] \times [0,2]$. If the curl is constant = 5, find the circulation.

$$\iint_R 5 \, dA, [0, 3] \times [0, 2]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 5 \cdot (3-0) \cdot (2-0) = 5 \cdot 3 \cdot 2 = 30.$$

Answer: $= 5 \cdot 3 \cdot 2 = 30$

17. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,3] \times [1,4]$. If the curl is constant = 4, find the circulation.

$$\iint_R 4 \, dA, [0, 3] \times [1, 4]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 4 \cdot (3-0) \cdot (4-1) = 4 \cdot 3 \cdot 3 = 36.$$

Answer: $= 4 \cdot 3 \cdot 3 = 36$

18. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,5] \times [0,3]$. If the curl is constant = 4, find the circulation.

$$\iint_R 4 \, dA, [0, 5] \times [0, 3]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 4 \cdot (5-0) \cdot (3-0) = 4 \cdot 5 \cdot 3 = 60.$$

Answer: $= 4 \cdot 5 \cdot 3 = 60$

19. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [0,5] \times [1,5]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [0, 5] \times [1, 5]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 1 \cdot (5-0) \cdot (5-1) = 1 \cdot 5 \cdot 4 = 20.$$

Answer: $= 1 \cdot 5 \cdot 4 = 20$

20. By Green's theorem, the circulation integral around a closed curve equals the double integral of curl over the enclosed region $R = [1,5] \times [1,3]$. If the curl is constant = 1, find the circulation.

$$\iint_R 1 \, dA, [1, 5] \times [1, 3]$$

→ By Green's theorem: circulation = integral integral (curl) dA .

$$\rightarrow = 1 \cdot (5-1) \cdot (3-1) = 1 \cdot 4 \cdot 2 = 8.$$

Answer: $= 1 \cdot 4 \cdot 2 = 8$



Work along a path

21. A vector field $F = (1, 3, 0)$ acts along a straight path with displacement vector $d = (2, 4, 0)$. Find the work done $W = F \cdot d$.

$$(1, 3, 0) \cdot (2, 4, 0)$$

$$\rightarrow W = F \cdot d = (1)(2) + (3)(4) + (0)(0) = 14.$$

Answer: **= 14**

22. A vector field $F = (0, -4, 3)$ acts along a straight path with displacement vector $d = (5, 4, 2)$. Find the work done $W = F \cdot d$.

$$(0, -4, 3) \cdot (5, 4, 2)$$

$$\rightarrow W = F \cdot d = (0)(5) + (-4)(4) + (3)(2) = -10.$$

Answer: **= - 10**

23. A vector field $F = (2, -1, 2)$ acts along a straight path with displacement vector $d = (6, 2, 2)$. Find the work done $W = F \cdot d$.

$$(2, -1, 2) \cdot (6, 2, 2)$$

$$\rightarrow W = F \cdot d = (2)(6) + (-1)(2) + (2)(2) = 14.$$

Answer: **= 14**

24. A vector field $F = (4, -2, 3)$ acts along a straight path with displacement vector $d = (5, 6, 1)$. Find the work done $W = F \cdot d$.

$$(4, -2, 3) \cdot (5, 6, 1)$$

$$\rightarrow W = F \cdot d = (4)(5) + (-2)(6) + (3)(1) = 11.$$

Answer: **= 11**

25. A vector field $F = (3, -5, 3)$ acts along a straight path with displacement vector $d = (6, 5, 1)$. Find the work done $W = F \cdot d$.

$$(3, -5, 3) \cdot (6, 5, 1)$$

$$\rightarrow W = F \cdot d = (3)(6) + (-5)(5) + (3)(1) = -4.$$

Answer: **= - 4**

26. A vector field $F = (-2, -4, 3)$ acts along a straight path with displacement vector $d = (1, 2, 1)$. Find the work done $W = F \cdot d$.

$$(-2, -4, 3) \cdot (1, 2, 1)$$

$$\rightarrow W = F \cdot d = (-2)(1) + (-4)(2) + (3)(1) = -7.$$

Answer: **= - 7**



27. A vector field $F = (4, -3, 0)$ acts along a straight path with displacement vector $d = (2, 4, 1)$. Find the work done $W = F \cdot d$.

$$(4, -3, 0) \cdot (2, 4, 1)$$

$$\rightarrow W = F \cdot d = (4)(2) + (-3)(4) + (0)(1) = -4.$$

Answer: $= -4$

28. A vector field $F = (1, -1, 2)$ acts along a straight path with displacement vector $d = (5, 2, 1)$. Find the work done $W = F \cdot d$.

$$(1, -1, 2) \cdot (5, 2, 1)$$

$$\rightarrow W = F \cdot d = (1)(5) + (-1)(2) + (2)(1) = 5.$$

Answer: $= 5$

29. A vector field $F = (2, 3, 0)$ acts along a straight path with displacement vector $d = (5, 5, 0)$. Find the work done $W = F \cdot d$.

$$(2, 3, 0) \cdot (5, 5, 0)$$

$$\rightarrow W = F \cdot d = (2)(5) + (3)(5) + (0)(0) = 25.$$

Answer: $= 25$

30. A vector field $F = (0, 3, 0)$ acts along a straight path with displacement vector $d = (3, 2, 0)$. Find the work done $W = F \cdot d$.

$$(0, 3, 0) \cdot (3, 2, 0)$$

$$\rightarrow W = F \cdot d = (0)(3) + (3)(2) + (0)(0) = 6.$$

Answer: $= 6$

